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Density Estimation Helps Adversarial Robustness

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***Abstract*—Adversarial attacks pose a threat to deep learning models, as they involve subtle disturbances that are imperceptible to human vision. In this paper, a classification network is introduced that also includes a density estimation head modeled using the decoder of a variational autoencoder. Incorporating the loss of the variational autoencoder during the training of the classifier aids in achieving a robust latent variable. The experimental findings show that the suggested model successfully defends against various gradient-based adversarial attacks, including FGSM, R-FGSM, MI-FGSM, and PGD, in both scenarios involving white-box and black-box contexts.**

***Index terms*—** Variational Autoencoder, Adversarial Robustness

I. Introduction

Deep networks have excelled in many different fields, yet it has been shown that they are susceptible to adversarial attacks. These attacks significantly degrade the performance of the networks by making slight adjustments to the input data, which are not perceptible to humans and are not anticipated to affect the predictions [1]. Particularly given the growing utilization of deep neural networks in fields like autonomous vehicles [2], developing methods to manage these sorts of inputs is more important than ever. When implementing deep learning models, it is crucial to have a system that is not just reliable and robust against attacks but also performs effectively.

Depending on how much information is accessible to the attacker, adversarial attacks may be classified as either white-box attacks or black-box attacks. In a white-box attack [3], the attacker has full access to and knowledge of the model being targeted, including its architecture and parameters. White-box attacks, which rely on having access to the model parameters, are generally more precise compared to black-box attacks [3]. Contrarily, black-box attacks prevent the adversary from accessing the model's internal workings, including its parameters or architecture, and also prevent them from sending queries to find out more about them.

To mitigate the effects of attacks coming from adversarial sources, several defensive mechanisms have been developed. A simple but powerful defense method, known as adversarial training, entails augmenting the training data of the classifier with adversarial samples to enhance its resistance to attacks [1], [4]. Another defense approach, known as network modifying, involves modifying the classifier's training procedure to minimize the magnitude of gradients [5]. Another sort of defense includes preprocessing input samples using modifications in order to eliminate adversarial noise before passing them on to a classifier [6], [7], [8], [9], [10]. These strategies may be used to resist white-box and black-box attacks, but not simultaneously [6], [11]. Certain defense mechanisms were created considering particular attack scenarios, making them susceptible to more recent methods of attack. The majority of current defense techniques do not adapt to all forms of adversarial attacks because, while they may prevent one type of attack, they may encounter another attacker who is aware of the defense mechanisms. Implementing such protection mechanisms can potentially result in performance overhead and diminish the model's predictive accuracy.

We offer a defensive mechanism in this research that can resist both white-box and black-box attacks. It employs a deep latent generative model known as a Variational AutoEncoder (VAE) to protect itself against adversarial attacks. To address the potential susceptibility of deep generative models to adversarial attacks, we incorporate a classifier during the training of the variational autoencoder to address this concern. This approach enables us to jointly optimize the training of the classifier and the deep generative model to improve the robustness of the model.

Following is the format for the remaining portions of the essay: A summary of relevant research on adversarial attacks, defensive measures, and background information on VAE is given in Section 2. In Section 3, we present the formal introduction of our defense mechanism. Section 4 presents the experimental results and comparisons with other defense strategies. The paper is concluded in Section 5.

II. Related Works

This work is related to three areas, which are explained in this section. Firstly, we delve into several attack models that have been used in the literature. Afterward, we review several defense strategies against these threats and discuss their advantages and disadvantages. Finally, we provide essential baseline information on VAE.

A. Adversarial Attacks

Recently, various techniques have been developed to generate adversarial samples, encompassing gradient-based, optimization-based, and model-based methods. Adversarial examples are crafted by applying perturbation steps that align with the adversarial gradients in gradient-based algorithms.

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In this study, our focus lies on gradient-based untargeted white-box attacks, specifically the Fast Gradient Sign Method (FGSM) [1], the Randomized Fast Gradient Sign Method (R-FGSM) [11], the Momentum Iterative Fast Gradient Sign Method (MI-FGSM) [12], and the powerful first-order method known as Projected Gradient Descent (PGD) [13]. We select these four techniques due to their representation of a wide range of attack methodologies, which enables us to thoroughly evaluate our defense strategy against them. In the subsequent sections, a detailed explanation of these attacks will be provided.

1) Fast Gradient Sign Method (FGSM) [1]: The generation of adversarial examples x^{adv} involves minimizing the probability associated with the true class y:

$$x^{adv} = x + \epsilon.sign(\nabla_x J(x, y)) \tag{1}$$

At every individual pixel of the input image, we move in the direction of the gradient of the cross-entropy loss $\nabla_x J(x, y)$. This is accomplished using the Fast Gradient Sign Method (FGSM), where ϵ represents the magnitude of change applied to each pixel, and the sign function is denoted by $sign()$. In essence, the FGSM updates the data in a single step by moving in the gradient ascending direction.

2) Randomized Fast Gradient Sign Method (R-FGSM) [11]: This attack is a straightforward but efficient way to enhance the effectiveness of FGSM against models that were trained adversarially. Before applying FGSM, a small random perturbation is introduced. Specifically, for $\alpha < \epsilon$, the original input x is modified by adding random noise as follows:

$$x^{'} = x + \alpha.sign(\mathcal{N}(0^d, 1^d)) \tag{2}$$

Subsequently, the FGSM attack is calculated on $x^{'}$, resulting in

$$x^{adv} = x + (\epsilon - \alpha).sign(\nabla_{x'} J(x^{'}, y)) \tag{3}$$

3) Momentum Iterative Fast Gradient Sign Method (MI-FGSM) [12]: It employs momentum to assist iterative gradient-based techniques in avoiding getting stuck in local maxima and to stabilize the update directions for perturbations, thereby further improving their attacking performance.

$$g_{t+1} = \mu.g_t + \frac{\nabla_x J(x_t^{adv}, y)}{\|\nabla_x J(x_t^{adv}, y)\|_1} \tag{4}$$

$$x_{t+1}^{adv} = Clip_x^{\epsilon}\{x_t^{adv} + \alpha.sign(g_{t+1})\} \tag{5}$$

where g_t stands for the cumulative sum of gradients collected throughout t iterations, with $g_0 = 0$ and μ representing the decay factor applied to the momentum term g_t, and $Clip_x^{\epsilon}\{\cdot\}$ is a function used to confine adversarial examples x_t^{adv} within a vicinity of x defined by the ϵ neighborhood.

4) Projected Gradient Decent Method (PGD) [13]: It is considered the preferred method for large-scale constrained optimization. The PGD attack, similar to the I-FGSM [14], iterates multiple times to search for an adversarial example while generating the attack. The authors argue that using noisy initial points results in a stronger attack compared to previous iterative approaches. By incorporating this stronger attack into adversarial training, they increase the effectiveness of their defense mechanism.

$$x_0^{adv} = x \tag{6}$$

$$x_{n+1}^{adv} = Clip_x^{\epsilon_p}\{x_n^{adv} + \alpha.sign(\nabla_x J(x_n^{adv}, y))\} \tag{7}$$

where α is step size and $Clip_x^{\epsilon_p}\{\cdot\}$ denotes clipping updated images to confine them inside the ϵ_p-ball of X. They repeat the investigation several times and assign the initial perturbation to a random location situated within the allowable ϵ-ball in order to avoid being trapped in a local minimum.

B. Defense Methods

In this work, we focus on purification techniques that utilize an auxiliary network to minimize adversarial perturbations. Purification aims to reduce or eliminate the possibility of adversarial attacks on the image.

MagNet [6] is a defense strategy that comprises some detector networks as well as a reformer network. The detector networks utilize the reconstruction error of the autoencoder to recognize whether a given example is adversarial or natural. Subsequently, the reformer network reconstructs adversarial instances to resemble natural examples while making minimal changes to the natural examples. The MagNet has demonstrated good performance in empirical tests against black-box attacks, but it is weak against white-box attacks. It performs poorly compared to Defense-GAN [7]. Rather than utilizing autoencoders, To lessen the effects of both white-box and black-box attacks, Defense-GAN [7] makes use of the powers of generative adversarial networks. The primary objective is to recognize the embedding space of the GAN model that contains the most accurate representation of the input image x. Utilizing the latent representation of the original picture, the modified image is produced, and obtained by training a generator on clean data, ensuring that it remains unaffected by adversarial distortions. Gradient descent optimization is employed to discover the optimal representation z, which minimizes the distance between the initial image x and the reconstructed image G(z). A drawback of Defense-GAN is its dependence on the performance and generative power of the GAN model. Additionally, Defense-GAN takes some time to attain its maximum defensive effectiveness due to its iterative nature. The PuVAE [8] maps an adversarial instance onto each class's manifold and chooses the nearest projection as a purified sample to remove adversarial distortions. PuVAE exhibits robustness against various attack methods even without prior knowledge of the attacks, and it does it with an inference time that is around 130 times quicker than Defense-GAN [7]. Defense-VAE [9] utilizes the VAE's capacity to comprehend the input data distribution to restore the possibly adversarial image to the manifold of the data. By feeding adversarial data to the VAE-encoder network in a forward-propagation approach, Defense-VAE can promptly identify a valid latent representation, denoted as z. The VAE decoder network is employed to produce a clean image from this latent representation. This method eliminates the need for expensive iterative online optimization. The reconstructed images not only have a faster overall creation time compared to Defense-GAN [7], but they are also more accurate. In [10], modifications are made to the training process of the Defense-VAE process, and additional loss functions are included, further enhancing the resistance of Defense-VAE [9] against adversarial attacks.

C. Variational Autoencoder

The Variational Autoencoder (VAE) [15] is a prominent deep generative model that utilizes latent variable models. It is composed of an encoder that converts the input image into a latent variable z and a decoder that reverses this conversion by converting the latent variable z back into the image domain.

$$z \sim Enc(x) = q(z|x) \tag{8}$$

$$x \sim Dec(z) = p(x|z) \tag{9}$$

As computing the maximum likelihood (ML) estimate for this latent variable model is impractical, an alternative is to optimize a variational lower bound (ELBO).

$$\begin{aligned} & L_{VAE} \\ & = E_{q(z|x)}[\log \frac{p(x|z)p(z)}{q(z|\hat{x})}] \\ & = E_{q(z|x)}[\log p(x|z) + D_{KL}(q(z|\hat{x})p(z))] \end{aligned} \tag{10}$$

where the first component denotes the reconstruction error and the second term acts as a regularizer to promote the posterior distribution's close resemblance to the prior distribution. In VAE, a common assumption is a simple unit Gaussian prior. In order to speed up calculation and make it possible to apply the reparameterization method, which reduces the variance of Monte Carlo sampling, a diagonal covariance Gaussian posterior is also assumed. VAE can produce high-quality images by using a generative model to imitate the distribution of the training images.

III. Proposed Method

A. Motivation

Although adversarial data may appear visually similar to real data, it can be confidently misclassified by a classifier. This phenomenon is particularly evident in image classification tasks, where adversarial images, despite their close resemblance to the original images in the image space, exhibit distinct characteristics in the feature space. Specifically, adversarial data points lie outside the manifold of the original data, commonly referred to as the off-manifold problem. As a result, when a neural network is applied to these images and features are extracted, there is a noticeable discrepancy between the feature representations of adversarial and original images, leading to misclassification. To solve this issue, one approach is to train a network that can learn a resistant manifold of images by extracting resilient features. This entails finding a set of features where the adversarial and original data points exhibit minimal differences in the feature space. To achieve this, generative models with hidden spaces are employed to extract these features and construct resistant manifolds.

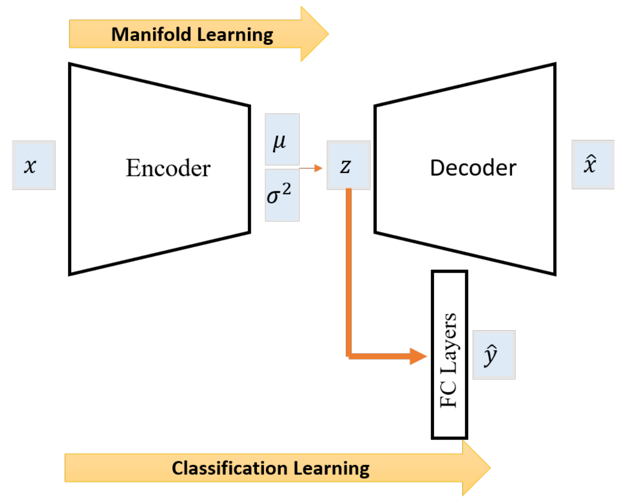


Fig. 1. The proposed model can be visually represented by three primary components: an encoder, a decoder, and a classifier. The classifier incorporates the encoding part and fully connected layers. The model operates through two primary pathways, namely manifold learning and categorized learning, which occur simultaneously.

Previous research on manifold-based approaches has utilized generative models to reconstruct adversarial images and subsequently incorporate these purified images into the classifier. While this approach enhances the model's resistance to attacks based on classifier parameters, it renders it vulnerable to attacks targeting the parameters of the generative model. In essence, generative models act as a black-box component, assuming that adversaries cannot access their parameters. However, this assumption is unrealistic since adversaries can access both the structure and parameters of the classifier, including the purifier model. This accessibility enables adversaries to design potent attacks, compromising the model's security, significantly reducing the classifier's accuracy, and disrupting the purification process performed by the generative model.

B. Our Method

We utilize the VAE model to acquire knowledge about the latent distribution of input data, which contains relatively abundant information. We propose combining the classifier and the VAE in a unified network. By incorporating the classifier into the model training procedure, we achieve a resistant manifold of data after the encoder, which can serve as a resistant feature extractor. A similar approach is suggested in [16] for situations in semi-supervised learning where unlabeled data are abundant but obtaining class labels is either costly or infeasible for the entire dataset. In this model, the latent code is effectively separated into two components, y and z, enabling both semi-supervised classification and controlled generation by keeping one of these components constant while generating the other. This approach breaks down the problem into two phases: first, an unsupervised pre-training phase using a Variational Autoencoder (VAE), and then a supervised learning phase involving a classifier, with separate training for each component. This method [16] is not suitable for the adversarial defense problem. However, our approach differs in that we incorporate the optimization of the classifier during training and merge these two phases. Our proposal is to employ a classifier to enhance the VAE's latent clustering capabilities while it is still in the training process. This method has not been extensively explored in previous research. As a result, we have chosen to combine both processes, in contrast to the approach recommended in [16].

This method is applicable to any classification task, and it is sufficient to consider the layers before the fully connected layer of the classifier as encoder layers. Similarly, the layers of the decoder can be determined simply as the inverse of the encoder layers. After the latent space, one or two fully connected layers are added for the classifier part. In other words, the encoding part, along with the fully connected layers, form our classifier part. The scheme of the introduced method can be seen in Figure 1, which includes two main paths: manifold learning and classified learning, occurring simultaneously. Manifold learning aims to access a resistant manifold through nonlinear transformations in the encoder, while classifier learning focuses on learning classification based on the robust hidden space.

In this method, the learning of the classifier and VAE is done simultaneously, and the loss function of this mechanism is computed as the combined sum of the classifier's loss function and the VAE's loss function. The training of the VAE involves minimizing a loss function that comprises two distinct objectives:

$$L_{total}(\theta, \phi, \beta) = L_{\beta VAE} + \alpha . L_{classification} \tag{11}$$

$$\begin{aligned} &L_{\beta VAE} \\ &\quad = E_{q(z|x)}[\log \frac{p(x|z)p(z)}{q(z|\hat{x})}] \\ &\quad = E_{q(z|x)}[\log p(x|z) + \beta D_{KL}(q(z|\hat{x})p(z))] \end{aligned} \tag{12}$$

$$L_{classification} = L_{CrossEntropy}(y, \hat{y}) = -\frac{1}{m}\sum_{i=1}^{m} y_i \log \hat{y_i} \tag{13}$$

The first objective, which corresponds to the VAE objective, is computed using equation 12. The subsequent objective, which measures the performance of the classifier using categorical cross-entropy, is determined by equation 13. The coefficients α and β are hyperparameters. Careful consideration must be given to choosing these hyperparameters, as a high value of this parameter does not guarantee an improvement in the accuracy of the classifier. There needs to be a compromise between this parameter and the parameter β, which is related to the Kolbeck-Liber loss, in order to achieve the best accuracy.

A constraint is introduced to the VAE to ensure that the classes are adequately separated from each other in the latent space, as well as to maintain a continuous latent space when optimizing the classifier, which is included in the VAE's loss function.

We have included adversarial training to improve the performance of the proposed method. In each learning stage, the original data is first attacked using the parameters of the encoder, followed by the fully connected layers. Subsequently, the attacked data is fed into the model, enabling it to estimate a more accurate manifold and extract more suitable features, ultimately leading to higher accuracy. Consequently, the loss function is indicated as below:

$$\begin{aligned} &L_{OURS} \\ &\quad = E_{q(z|x_{adv})}[\log \frac{p(x_{adv}|z)p(z)}{q(z|\hat{x})}] \\ &\quad = E_{q(z|x_{adv})}[\log p(x_{adv}|z) + \beta D_{KL}(q(z|\hat{x})p(z))] \end{aligned} \tag{14}$$

IV. Experiments

We regard both white-box and black-box attacks to assess the effectiveness of our method in defending against them. To generate adversarial examples, we utilize gradient-based attacks such as FGSM, Rand-FGSM, MI-FGSM, and PGD. We performed our investigation on the MNIST dataset [17]. The VAE structure used for single-channel data, specifically MNIST, comprises several convolution layers, which are detailed in Figure 2.

During the training process, we employ adversarial training to enhance the robustness of the model against adversaries. We generate adversarial examples for each attack individually. At each stage, adversarial data is generated based on the classifier section and then merged with the original data. For training our proposed model, we set the maximum adversarial perturbation to 0.3 and train our proposed model using the adversarially generated training dataset for 10 cycles until reaching convergence. Typically, our model converges within 5 cycles. We utilize the Adam optimizer with a beginning learning rate of 0.001, which exponentially drops to around 0.00001. The values of the hyperparameters are chosen as follows: hidden space size = 128, α = 0.85, and β = 1.

A. Accuracy Evaluation

For the MNIST dataset, the accuracy value for the original, non-attacked data is 99.03, indicating that our method functions proficiently on clean data. In this part, our technique is compared with two previous state-of-the-art approaches in both scenarios involving white-box and black-box contexts.

1) Results on White-Box Attacks: White-box attacks rely on the gradients of the original model and are predicated on the assumption that the adversary has full knowledge of the model's parameters. In this research, the performance of the introduced method is compared to two state-of-the-art methods: the adversarial training method and the Defense-VAE method.

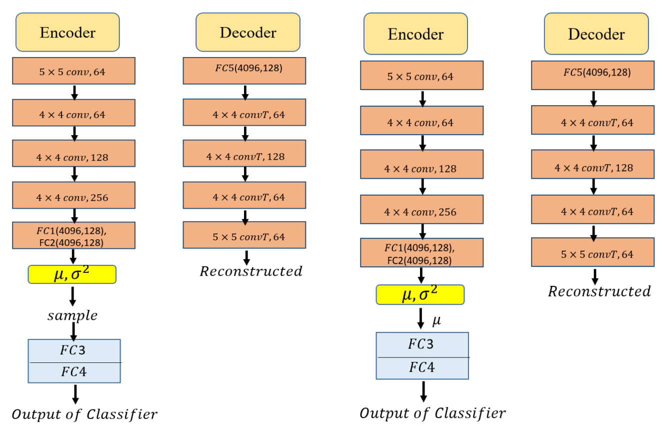
In Table I, the accuracy values for white-box attacks are reported, and it is noticeable that the introduced method is more successful against adversarial attacks compared to the other two methods. The difference between the introduced method and the Defense-VAE is not significant, which may be attributed to the simplicity of the MNIST dataset.

TABLE I
CLASSIFICATION ACCURACY OF SEVERAL DEFENSIVE STRATEGIES AGAINST FGSM, RAND-FGSM, MI-FGSM, AND PGD WHITE-BOX ATTACKS ON THE MNIST DATASET.

Attack Method	No Attack	No Defense	Adv. Training	Defense VAE	Ours
FGSM	99.03	13.6	69.8	96.75	99.5
R-FGSM	99.03	11.9	62.6	97.34	98.77
MI-FGSM	99.03	0.12	42.5	94.3	95.0
PGD	99.03	0.0	42.9	94.9	95.4

TABLE II
CLASSIFICATION ACCURACY OF SEVERAL DEFENSIVE STRATEGIES AGAINST FGSM, RAND-FGSM, MI-FGSM, AND PGD BLACK-BOX ATTACKS ON THE MNIST DATASET.

Attack Method	No Attack	No Defense	Adv. Training	Defense VAE	Ours
FGSM	99.03	21.59	81.1	92.3	94.4
R-FGSM	99.03	20.29	81.9	92.5	94.6
MI-FGSM	99.03	0.12	80.6	93.0	96.6
PGD	99.03	0.0	80.0	93.7	97.8



(a) Train (b) Test

Fig. 2. The architecture of encoder and decoder of VAE in train and test phase. During the test phase, only μ is injected into the fully connected layers, and σ^2 is integrated out.

TABLE III
THE ARCHITECTURE USED FOR BLACK-BOX ATTACKS' SUBSTITUTION MODEL

Layers
Conv(*,32,3,1,1) + ReLU
Conv(32,32,3,1,1) + ReLU
MaxPooling2d(2)
Conv(32,64,3,1,1) + ReLU
Conv(64,64,3,1,1) + ReLU
MaxPooling2d(2)
FC(3136, 200) + ReLU
FC(200, 10) + ReLU
Softmax

2) Results on Black-Box Attacks: The transferability of adversarial attacks is a fundamental characteristic, making it essential to evaluate the model's resistance to black-box attacks, which happen when the attacker does not have knowledge about the architecture of the model. The architecture employed for black-box attacks is presented in Table III.

Table II presents the accuracy of black-box attacks on the MNIST data. As observed, the introduced defense method against iterative attacks (such as MI-FGSM and PGD) exhibits higher resistance ability, which is also evident in the Defense-VAE method. Consequently, it can be concluded that methods utilizing variational autoencoders tend to offer relatively greater resistance against iterative black-box attacks.

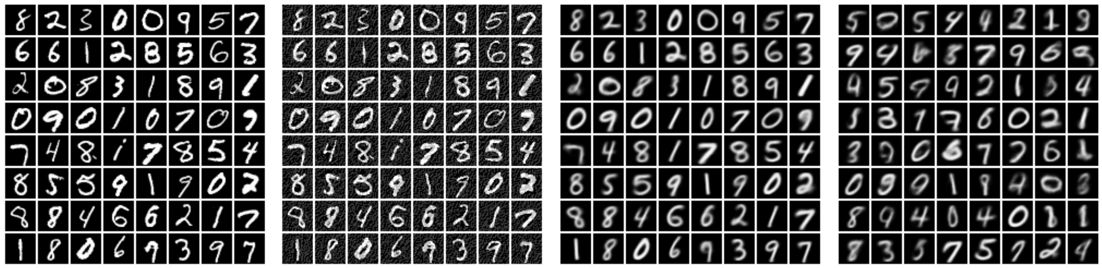
B. Visualization

Figure 3 presents examples of original data, adversarial samples, reconstruction images, and generated samples. The close resemblance between the reconstructed and original images demonstrates the method's effectiveness in reconstructing original data from adversarial samples while boosting the model's resistance to adversarial attacks. Additionally, the accurate generation of samples supports the applicability of this method in semi-supervised settings with limited labeled data and adversarial data. By successfully reconstructing and generating plausible images, this method enhances both the classification network and the generative model.

V. CONCLUSION

The innovation of this research lies in enhancing classification resistance by utilizing the hidden space efficiency of the Variational Autoencoder (VAE) through simultaneous training of the classifier and VAE. Moreover, the introduced defense method demonstrates adaptability to multiple attack types, in contrast to the limitations of previous approaches. Notably, the introduced method offers a clear understanding of the underlying reasons for its resistance, substantiated by a comprehensive background. By examining the outcomes of the introduced approach that incorporates a variational autoencoder, we can draw the following inferences.

- The learning process maintains the accuracy of clean data without a decrease, which ensures reliable classification results.
- The Defense-VAE method is highly susceptible to attacks on the encoder part, and all the compared results were obtained under the assumption that the classifier part was targeted. In contrast, our method specifically targets



(a) (b) (c) (d)

Fig. 3. Examples from MNIST under R-FGSM attack. (a) Original images, (b) adversarial images, (c) reconstructed images, and (d) generated samples

both the encoder part and all fully connected layers simultaneously. Despite this more challenging scenario, the introduced method has shown superior performance. Notably, the introduced method is effective regardless of which parts of the network the adversary can access.

- Sharing parameters between discriminative and generative models not only reduces the overall parameter count but also saves time during the training process.
- The introduced method produces reconstructed and generated images suitable for enhancing semi-supervised learning approaches, expanding its potential applications beyond classification tasks.

Overall, this research introduces a novel approach that leverages the advantages of a variational autoencoder to enhance classification resistance, demonstrating adaptability to various attack types and providing valuable insights into its effectiveness and potential applications.

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